Course notes

Throughout this chapter, coordinates are given in an orthonormal frame of space. This assumption is necessary to identify a normal vector from the coefficients of an equation and to calculate distances.

Parametrised lines

The line through $A(x_A, y_A, z_A)$ with non-zero direction vector (a, b, c) has the representation

$$d : \begin{cases} x = x_A + at \\ y = y_A + bt \\ z = z_A + ct \end{cases} \qquad (t \in \mathbb{R}).$$

A point belongs to the line if there exists *one and the same* real number t satisfying the three equalities. A line has many representations: change the starting point or multiply the direction vector by a non-zero real number.

Example. $(x, y, z) = (1, 2, 0) + t(2, -1, 3)$ describes the same line as $(3, 1, 3) + s(4, -2, 6)$; the link is $t = 1 + 2s$.

Cartesian planes

> **Key point · Equation of a plane.** If $(a, b, c) \neq (0, 0, 0)$, the set $ax + by + cz + d = 0$ is a plane with normal vector (a, b, c). The plane with normal $\vec{n}$ through A is constructed by writing $\overrightarrow{AM} \cdot \vec{n} = 0$.

Proof. The condition $\overrightarrow{AM} \cdot \vec{n} = 0$ is written $a(x - x_A) + b(y - y_A) + c(z - z_A) = 0$. Expanding, $d = -ax_A - by_A - cz_A$. Conversely, every point satisfying this equation satisfies the orthogonality condition, hence belongs to the plane.

Example. The plane with normal $(2, -1, 3)$ through $A(1, 2, 0)$ is $2(x - 1) - (y - 2) + 3z = 0$, that is $2x - y + 3z = 0$.

Intersections

For a line and a plane, replace x, y, z in the equation of the plane by their parametric expressions. A solution in t gives a single intersection point; an identity gives a line contained in the plane; a contradiction gives a parallel line that is disjoint from the plane. For two planes, non-collinear normals give an intersection along a line; solve the system of two equations by choosing one variable as a parameter. If the normals are collinear, the planes are parallel: distinguish coincident from disjoint. For two lines, use independent parameters s, t and solve the three equations. With no solution and non-collinear direction vectors, the lines are not coplanar.

Projection onto a plane

If $\mathcal{P} : ax + by + cz + d = 0$, look for the orthogonal projection of M onto the normal line through M: $H = M + t(a, b, c)$. Substitute into the equation of the plane to determine t, then calculate MH. We deduce, without making it a formula to be learned in isolation,

$$d(M, \mathcal{P}) = \frac{|ax_M + by_M + cz_M + d|}{\sqrt{a^2 + b^2 + c^2}}.$$

Essential check. After calculating a projection, check simultaneously that $H \in \mathcal{P}$ and that $\overrightarrow{MH}$ is collinear with the normal.